

Data Card

Latent Probing for Toxicity Detection

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Team: nyriagongo | Programme: AI Saturdays Lagos, Cohort 10

Dataset summary

The project builds a harmonized binary toxicity corpus from public research datasets. Extraction produced 149,528 records from 13 sources. Cleaning, normalization, filtering, and cross-source deduplication retained 108,468 examples from 12 sources across nine dataset families.

Prepared source counts were Civil Comments 26,218; Twitter toxicity/offensiveness corpora 27,572; Wikipedia talk-page attacks 11,314; Berkeley D-Lab hate speech 16,000; RealToxicityPrompts 14,000; Aegis Safety 6,152; Offensive Language 3,496; HateCheck 2,326; and ToxicChat 1,390.

Counts describe the reproducible prepared corpus used by this project and may differ from the full upstream datasets because of caps, filtering, unavailable rows, or deduplication.

Intended task and labels

The intended task is binary text classification. Label 1 means the upstream annotation or policy mapping indicates toxic, hostile, hateful, threatening, abusive, or offensive content. Label 0 means the mapped example is non-toxic for this task.

Because upstream taxonomies differ, the binary mapping compresses distinctions such as hate speech, insult, threat, profanity, harassment, and unsafe conversational content. It must not be interpreted as a universal definition of toxicity.

Collection and processing

The repository's extraction pipeline retrieves or reconstructs datasets from their permitted public sources, maps schemas to a common text-and-label format, removes unusable rows, normalizes fields, and deduplicates records. Fixed seeds and source identifiers support repeatability and source-aware evaluation.

Inputs are tokenized with the Gemma tokenizer and truncated to 64 tokens. The released submission does not include raw activation matrices or the full text corpus. Those artifacts are large, may contain harmful language, and can have redistribution limits.

Splits and evaluation

Competition development and testing sets are evaluated by CodaBench using accuracy. For internal robustness checks, source membership is retained and leave-one-source-out evaluation tests transfer to a held-out data source. The pooled transfer result was accuracy 0.795946 and AUROC 0.832204, lower than the final testing score and therefore an important caution against assuming uniform domain performance.

- Final probe training: 108,468 prepared rows; standardization and one LDA head are fitted once.
- Internal validation: nine family-held-out folds; each selection training fold is capped at 40,000 rows.
- CodaBench development: 1,700 rows used to compare packaged candidates, not to fit probe coefficients.
- CodaBench testing: 1,360 rows used only for the final held-out score.

Model and training configuration

The frozen backbone is `google/gemma-2-2b`, a 26-block decoder-only Transformer with hidden width 2,304. PyTorch and Hugging Face Transformers extract the masked-mean representation at hidden state index 14 after truncation to 64 tokens, using extraction batches of 64. The classifier is one standardized shrinkage-LDA head with λ 0.6 and a top-12/17 rank quota.

The backbone receives no weight updates. LDA uses a closed-form covariance estimate and linear solve; therefore optimizer, learning rate, and gradient-training epochs are not applicable. Model selection compares five LDA shrinkage values (0.02, 0.1, 0.3, 0.6, and 0.9) and other registered linear probes before one final refit.

Known risks and limitations

- Toxicity is subjective and policy-dependent; annotators may disagree.
- Identity terms can become spurious shortcuts and create false positives.
- English-heavy sources under-represent many languages and cultural contexts.
- Platform-specific language, sampling, and annotation procedures cause domain shift.
- A binary label hides severity, target, intent, quotation, counterspeech, and context.
- Truncation can remove relevant context from long messages.
- Public datasets may contain personal information or distressing content despite upstream controls.

Access, licensing, and governance

Users must follow every upstream dataset's licence, terms, and access conditions. Raw datasets are not redistributed in this repository. Where access permits, scripts automate retrieval; otherwise the README documents the required upstream step. Gemma access is governed by its model licence and may require the user to accept terms through Hugging Face.

Do not commit access tokens, local caches, raw text exports, or embeddings. Restrict access to sensitive working data, minimize retention, and delete caches when no longer required.

Recommended use

Use this corpus for research, benchmarking, and supervised moderation support with human review. Before operational use, evaluate local data, disaggregated error rates, dialect and identity-term slices, emerging vocabulary, and false-positive/false-negative costs. Provide an appeals path and document threshold changes.

Maintenance

Record upstream versions, extraction date, checksum, mapping changes, and known exclusions for each rebuild. Re-run source-aware tests when the corpus, tokenizer, model, or moderation policy changes. Treat drift and stakeholder feedback as reasons to revise the dataset rather than merely recalibrate a threshold.

Team and contacts

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All four team contributors receive equal contribution credit. Hamna is also in another team; her role with team nyriagongo covered shared research and Cohort Challenge work and does not imply exclusive membership.